

Mode Choice and Policy Welfare in Jabodetabek

An MNL → NL → MXL Comparison

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Motivation

- Jabodetabek: 30M+ population, 3.5M daily commuters — car/moto-dominated
- Transit expansion is spatially uneven: J1b (Parung) and J3b (Gading Serpong) have **no formal transit** at all
- **Which policies maximize welfare? How are gains distributed?**

Research framework: 6-mode choice set → MNL/NL/MXL → AIC/LR/Wald → best model → logsum ΔCS → 8 policy scenarios A–H

Data: 5,000 synthetic persons (NL GEV DGP, $\lambda=0.70$) + real r5py GTFS transit LOS + Ilahi et al. (2021) β anchors

Study Area & Data

Feature	Detail
Zones (7)	J1a–J5, Bodetabek → Jakarta CBD
Modes (6)	Car, Motorcycle, KRL, TransJakarta, RoyalTrans, MRT
Nest structure	Transit {KRL, MRT, TJ, Royal} · Private {Car, Moto}
Population	5,000 synthetic; 3 income segments (BPS Susenas 2023)
Transit LOS	r5py GTFS routing (AM peak 07:00–09:00, fallback 180 min)

MNL: Specification & IIA Problem

$$V_{mn} = ASC_m + \beta_{\text{time},m} \cdot T_{zn} + \beta_{\text{cost}} \cdot C_{zn}$$

12 parameters · L-BFGS-B MLE · KRL = reference alternative

Result	Value
Parameter recovery	12/12 within 2 SE
ρ^2 / AIC	0.280 / 10,121.65

IIA is violated. Red-bus/blue-bus: clone KRL as "KRL Express" → MNL nearly doubles transit share (11.7% → 21.8%). NL cross-elasticity: TJ + 0.1 utility → transit nest loses 1.90 pp, private nest loses 4.14 pp. Which is not what we want. **4.67** → 1.47. **4.67** → 1.47.

Nested Logit: $\lambda = 0.763$

Two nests · 13 parameters · FIML via L-BFGS-B

$$P_n(m) = P_n(m|k) \cdot P_n(k), \quad P_n(k) = \exp(\lambda I_{kn}) / \sum_{\ell} \exp(\lambda I_{\ell n})$$

Parameter	Estimate
λ	0.763 ± 0.068
95% CI	[0.627, 0.900] — excludes 1.0
LR test vs MNL	$\chi^2 = 8.57$, df = 1, p = 0.003
Recovery	13/13 within 2 SE
AIC / BIC	10,115.08 / 10,199.80

Mixed Logit: $\hat{\sigma} = 0.010$ (n.s.)

Random $\beta_{\text{cost}} \sim N(\mu, \sigma) \cdot R = 100$ Halton draws (base=2)

$H_0: \sigma_{\text{cost}} = 0$ — any heterogeneity beyond NL nest correlation?

Parameter	Estimate	SE
σ_{cost}	0.010	0.033
μ_{cost}	-0.037	0.183

Wald: $W = 0.091$, **p = 0.763** → fail to reject H_0

Positive control: Mixed-DGP ($\sigma_{\text{true}}=0.02$) → Wald $p \approx 0$ ✓

Model Selection

Criterion	MNL	NL	MXL
K	12	13	13
LL(β)	-5,048.83	-5,044.54	-5,048.79
AIC	10,121.65	10,115.08	10,123.59
BIC	10,199.86	10,199.80	10,208.31
LR vs MNL (p)	—	0.003	0.799
Wald $\sigma=0$ (p)	—	—	0.763

NL beats MNL by $\Delta AIC = -6.6$; beats MXL by $\Delta AIC = -8.5$. LR rejects H_0 at $p < 0.01$. Wald fails to reject $\sigma=0$. **NL selected** as the

Logsum Welfare Measurement

$$EMU_n = \ln \sum_k \exp(\lambda \cdot I_{kn}), \quad I_{kn} = \ln \sum_{j \in k} \exp(V_{jn}/\lambda)$$

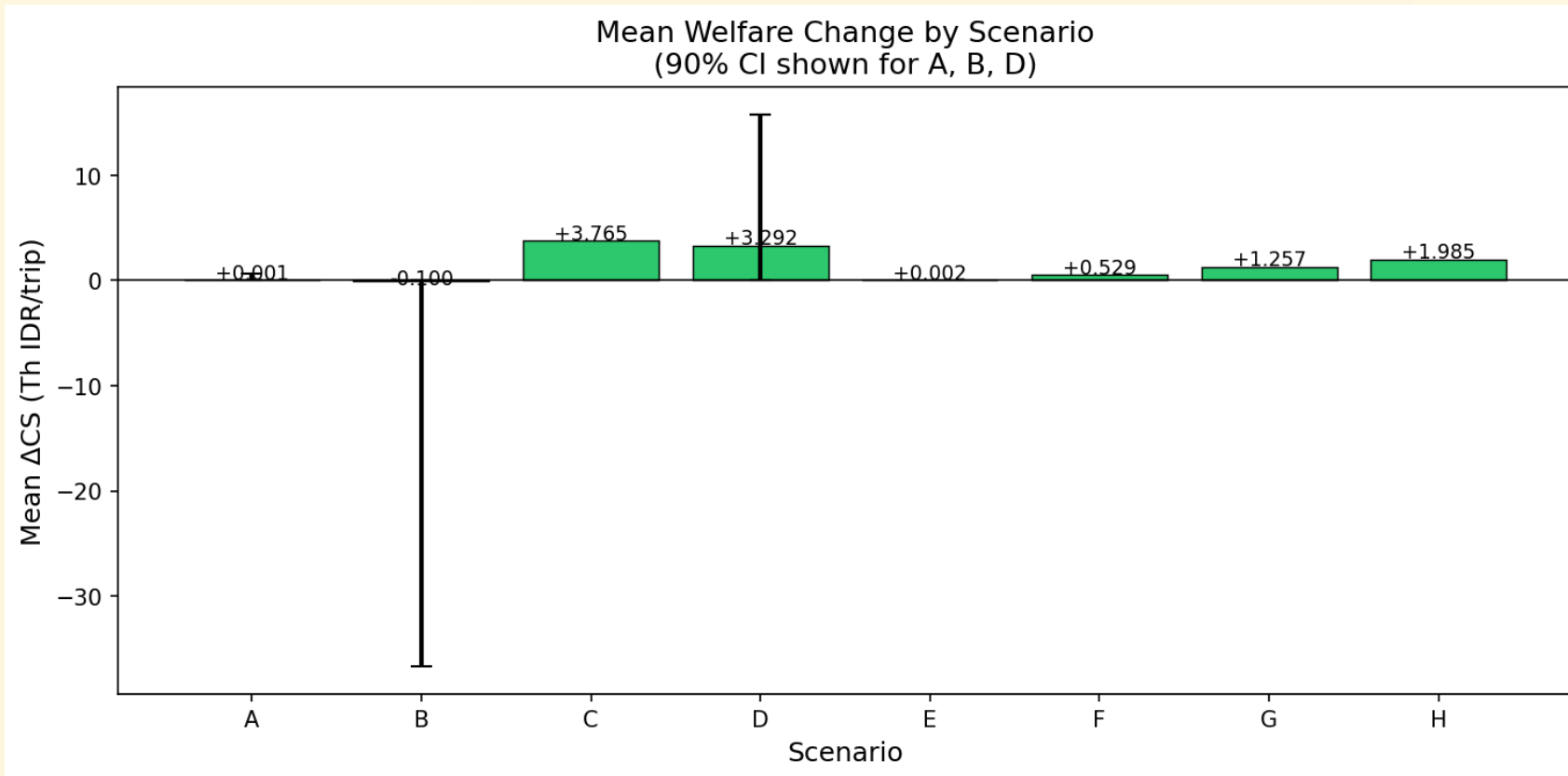
$$CS_n = \frac{EMU_n}{|\beta_{\text{cost}}|}, \quad \Delta CS_n = \frac{EMU_n^{\text{policy}} - EMU_n^{\text{baseline}}}{|\beta_{\text{cost}}|}$$

$$\Delta W_{\text{annual}} = \sum_n \Delta CS_n \times 250$$

Baseline CS: -53.65 Th IDR/trip (P10: -115.30, P90: -9.48)

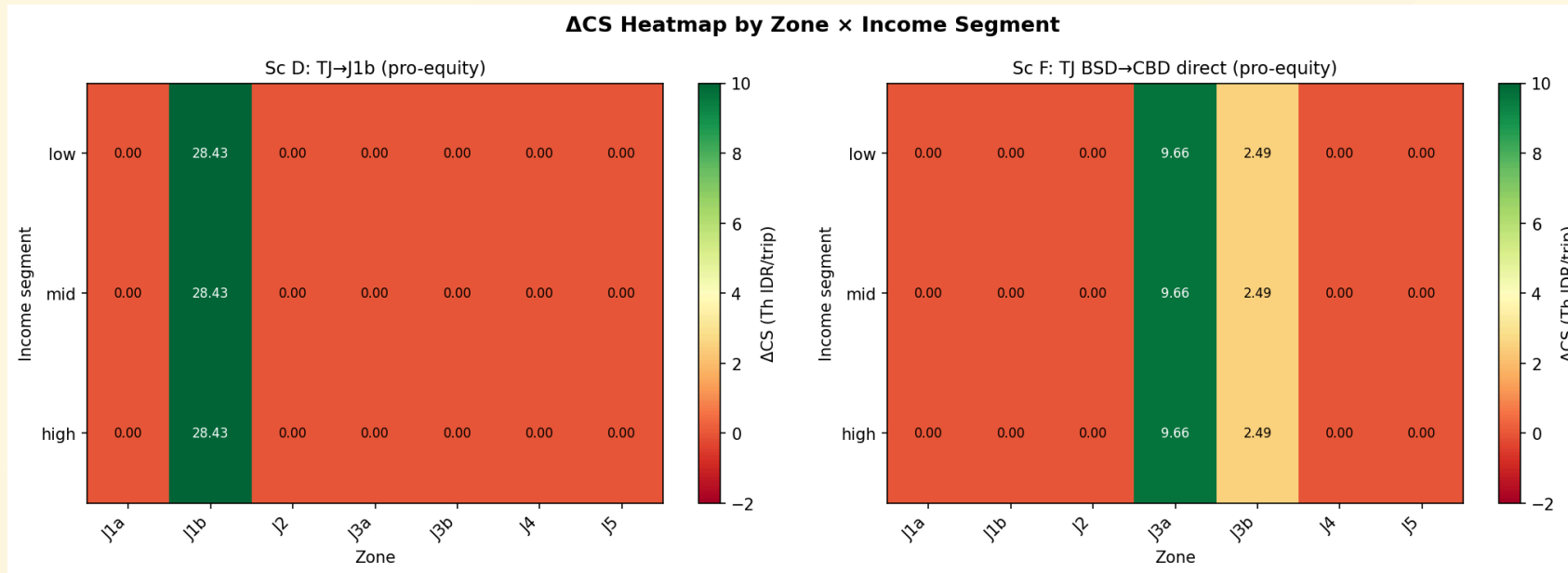
$\beta_{\text{cost}} = -0.077$ (SE = 0.097) · Bootstrap CIs use truncated-Normal
draws (upper bound = 0.3 | β_{cost} |)

Policy Scenarios – Overview



Top 3 by ΔCS : **C** (KRL freq, +3.76) > **D** (TJ $\rightarrow$ J1b, +3.29) > **H**
(RovalTrans fare, +1.99)

Policy Scenarios – Distributional Impact



Sc C maximizes aggregate welfare but J1b/J3b receive zero. **Sc D** concentrates gains in transit-orphan J1b (+28.43 Th IDR). **Sc F** benefits both J3a (+9.66) and J3b (+2.49) through route restructuring. Budget transit (TJ at Rp 3,500) draws from motorcycles

Conclusions & Limitations

Conclusions:

1. NL is the appropriate model: $\lambda=0.763$, LR $p=0.003$, MXL adds no signal (Wald $p=0.763$)
2. Top aggregate gain: Sc C (KRL freq, +3.76 Th IDR/trip, +6,580 Bn/yr)
3. Most pro-equity: Sc D (TJ $\rightarrow$ J1b, +3.29) — concentrated in the most underserved zone
4. **Bundle** service quality improvements with network expansion

Limitations: Synthetic DGP. Car share $\sim 1\%$ (full economic cost vs

Thank you
Questions?

Backup A: Why $\sigma = 25$?

Scale identification theorem (Train 2009 §3.7):

True utility: $U = V + \varepsilon, \varepsilon \sim \text{Gumbel}(0, \mu) \rightarrow$ Estimated: $\tilde{U} = \tilde{V} + \tilde{\varepsilon}, \tilde{\varepsilon} \sim \text{Gumbel}(0, 1)$

Normalization compresses all coefficients: $\tilde{\beta} = \beta / \mu$

VOT = $\beta_{\text{time}} / \beta_{\text{cost}}$ is preserved — μ cancels. β magnitudes differ from Ilahi's, but the behavioral metric (VOT) recovers exactly.

Without $\mu=25$, utility gaps are too large and choice becomes deterministic (99%+ choosing one mode). With $\mu=25$, β estimates have larger SEs (the trade-off), but the Gumbel noise produces

Backup B: Why Drop Ride-Hailing & LRT?

Mode	Reason for exclusion
2WRH (GoRide/GrabBike)	Ilahi $\beta_{\text{time_2wrh}} = -5.10$ estimated on short urban trips $\rightarrow$ near-zero share on 30–105 km commutes. Parameter non-transferability (Wardman et al. 2016).
4WRH (GoCar/GrabCar)	ASC calibration required $>+40$ utility units to match BPS aggregate share — outside defensible range.
LRT Jabodebek	Available in only 1 of 7 zones (J2). Single-zone presence $\rightarrow$ ASC absorbs zone

Backup C: Why Car Share ~1%?

LOS skim uses **full economic cost**: toll + fuel + parking $\approx$ Rp 130,000 per trip from outer zones.

Commuters perceive **marginal cost** (fuel $\approx$ Rp 30,000) once the car is owned — the fixed ownership cost is sunk at trip level.

With β_{cost} anchored to Ilahi (2021), full-cost specification suppresses Car utility. We keep the conservative specification to preserve the Ilahi anchor rather than introducing an unvalidated cost adjustment.

Trans-ESG Final Project, Hiroshima University, April 2024

Result: 51 of 5,000 persons choose Car. $\beta_{\text{time,car}}$ hits zero bound